

Petalino Biologino and the Lipid Pack Rescue 花瓣仔與脂質背包救援

I'll pack later!
遲啲先執啦!

FAT PACK
脂肪背包

Empty Fat Pack
is NOT a joke!
空背包唔係搞笑!

FAT PACK
脂肪背包

BIO
CAMP

Night hike begins...
夜行開始...

So cold...
好凍...

No lipids = no fuel,
warmth, or protection!
冇脂質 = 冇能量、保暖、保護!

FAT PACK
脂肪背包

1
FUEL
能量



2
MEMBRANE
細胞膜



3
SIGNAL
信號



I'll build it
properly!
我會認真砌好!

1

STRUCTURE
GATESTRUCTURE GATE / 結構關
TRIGLYCERIDE TEAM / 甘油三酯小隊Meet
Glycerol!
認識甘油!GLYCEROL
甘油GLYCEROL
= 1
甘油 = 1

2

Three tails.
Count them!
三條尾，數清楚!

FA1

FA2

FA3

FATTY ACIDS
= 3
脂肪酸 = 3

3

Need **THREE**
fatty acids!
要三個脂肪酸!

FAIL

4

GLYCEROL
甘油3 FATTY ACIDS
3 脂肪酸TRIGLYCERIDE
甘油三酯3 WATER
3 水GLYCEROL
甘油3 FATTY ACIDS
3 脂肪酸TRIGLYCERIDE
甘油三酯3 WATER
3 水CONDENSATION = form bonds + release water
縮合 = 成鍵 + 釋出水Slot 1
filled!
第一格滿咗!

1

Triglyceride Jobs / 甘油三酯嘅工作

Triglycerides
store energy
for later.

甘油三酯儲能量
留待後用。



ENERGY STORAGE
能量儲存

FAT PACK
脂肪背包

2

Lipids keep
body heat in.

脂質幫身體
留住熱能。



INSULATION 保溫

FAT PACK
脂肪背包

3

Fat cushions and
protects organs.

脂肪緩衝保護器官。

BOING!

ORGAN PROTECTION 器官保護

FAT PACK
脂肪背包

4

See? Prep
matters.
睇到未?
準備有用!

FAT PACK
脂肪背包

1
STORE
儲能



2
INSULATE
保溫



3
PROTECT
護器官



Two slots
still empty!
仲有兩格空!

★ Triglycerides = store · insulate · protect / 儲能 · 保溫 · 護器官 ★

Membrane Gate / 細胞膜關

Phospholipid Crew / 磷脂質小隊

The gate needs a wall!
呢度要一堵牆!

CELL GATE
細胞關

COACH
BEAN

Meet PHOS / 磷頭仔!

PHOSPHATE
HEAD
磷酸頭

GLYCEROL
BODY
甘油身體



FATTY ACID
TAILS (2)
脂肪酸尾巴 (2條)

Phospholipid =
phosphate + glycerol +
2 fatty acids

磷脂質 =
磷酸 + 甘油 + 2 脂肪酸

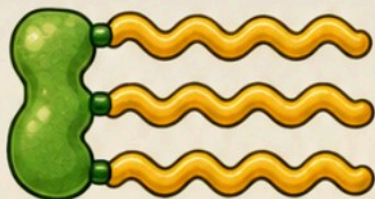
Two tails,
plus a phosphate
head!

兩條尾，
加磷酸頭!

COACH
BEAN

Triglyceride / 甘油三酯

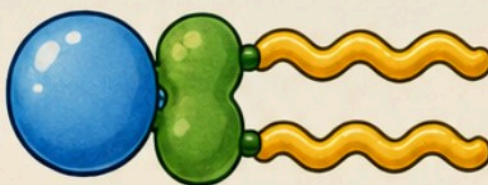
Glycerol + 3 fatty acids
= energy storage
甘油 + 3 脂肪酸 = 儲存能量



3 tails
(3 條尾)
No phosphate
head
(無磷酸頭)

Phospholipid / 磷脂質

Phosphate + glycerol + 2 fatty acids
= membrane building
磷酸 + 甘油 + 2 脂肪酸 = 構建細胞膜



2 tails
(2 條尾)
Plus phosphate
head
(有磷酸頭)

Phospholipids =
main component of
cell membranes

磷脂質 =
細胞膜主要成分

Phospholipids line up
head-out / tails-in
形成雙層牆壁!

CELL GATE
細胞關

Slot 2
filled!
第二格
滿咗!

FAT PACK / 脂肪背包

1
FUEL
能量



2
MEMBRANE
細胞膜



3
SIGNAL
信號



Great work,
crew!
做得好，團隊!

COACH
BEAN

1

SIGNAL GATE + EMERGENCY / 信號關 + 緊急狀況

Who can help stabilize and send signals?
邊個可以穩定同埋發送信號?



STERI / 固醇仔

I'm STERI!
I stabilize and send signals!
我係固醇仔!
我穩定同埋發送信號!



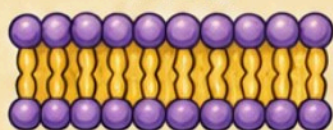
2

Steroids (e.g. cholesterol) → sex hormones + cell membranes

類固醇 (如膽固醇) → 性荷爾蒙 + 細胞膜

A CELL MEMBRANES

細胞膜



Helps membranes stay stable.

幫助細胞膜保持穩定。

B SEX HORMONES

性荷爾蒙



Messenger badges in the body.

身體入面嘅信使徽章。

Learn it.
No shortcuts!
記住，冇捷徑!



FAT PACK PROGRESS

- 1 FUEL ☒
- 2 MEMBRANE ☒
- 3 SIGNAL ☒

3

FINAL CLIMB! / 最後一關!

Emergency!
Storage locked tight!
We need help!
緊急狀況!
儲存被鎖住咗!
需要幫手!



HYDROLYSIS / 水解反應

H₂O SPLASH SQUAD 水滴救援隊

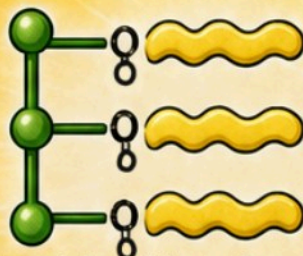


Water:
I can help!
我可以幫手!

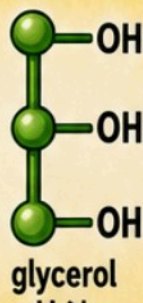


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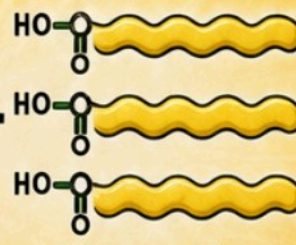
HYDROLYSIS IN ACTION! / 水解反應進行中!



triglyceride + 3 water
甘油三酯 + 3 水



glycerol
甘油



3 fatty acids
3 脂肪酸

triglyceride + 3 water → glycerol + 3 fatty acids
甘油三酯 + 3 水 → 甘油 + 3 脂肪酸

HYDROLYSIS = break bonds + use water
水解 = 拆鍵 + 用水

Now use what you built.
Move!
而家用你砌好嘅，走!



1

WARM CAMPFIRE / 暖營火

The Lipid Pack
is complete!
脂質背包齊晒!



2

I wasn't
prepared...
I get it now.
我之
前
冇
準備...
而家明啦!



Next time,
pack before
you brag.
下次未帶口之前，
先做好背包。



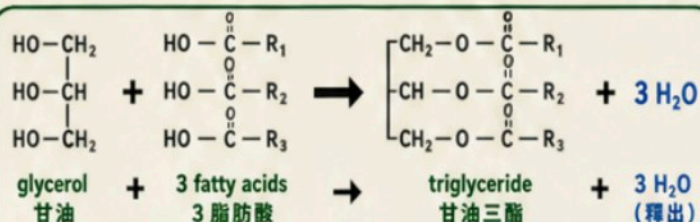
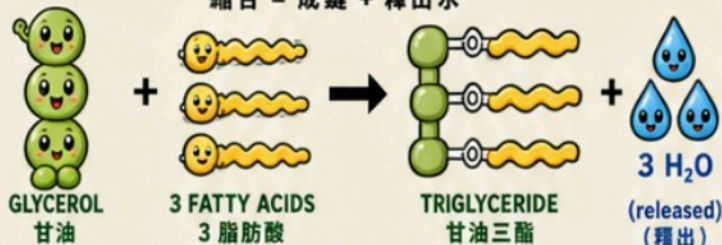
Build it to store. Break it to use.
Wall it to protect. Signal it to control.

砌起嚟儲。拆開嚟用。
砌牆嚟護。發信號嚟調控。

3A

CONDENSATION / 縮合

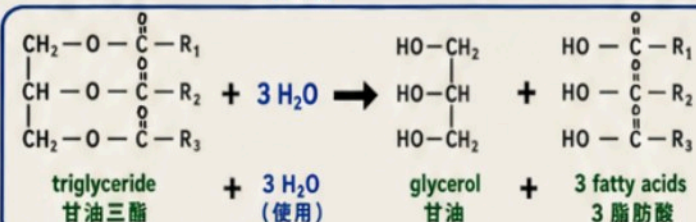
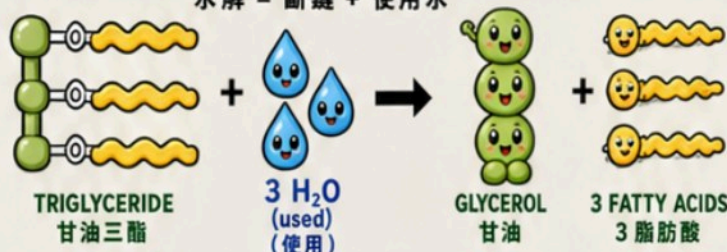
Form bonds + release water
縮合 = 成鍵 + 釋出水



3B

HYDROLYSIS / 水解

Break bonds + use water
水解 = 斷鍵 + 使用水



1

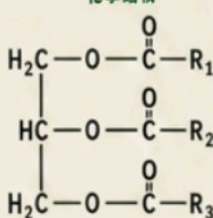
TRIGLYCERIDES / 甘油三酯

CARTOON

Glycerol 甘油

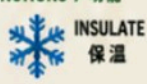


CHEMICAL STRUCTURE 化學結構



1 glycerol + 3 fatty acids
1 甘油 + 3 脂肪酸

FUNCTIONS / 功能

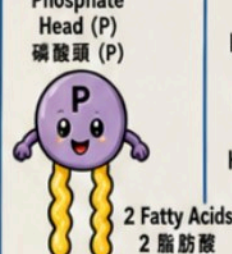


2

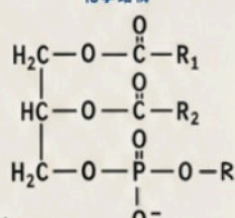
PHOSPHOLIPIDS / 磷脂質

CARTOON

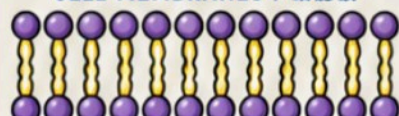
Phosphate Head (P) 磷酸頭 (P)



CHEMICAL STRUCTURE 化學結構



CELL MEMBRANES / 細胞膜



3

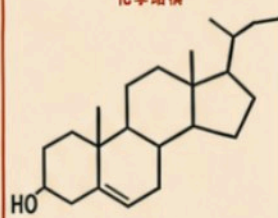
STERIODS / 類固醇

CARTOON

Steri



CHEMICAL STRUCTURE 化學結構



FUNCTIONS / 功能

